# Where we started: the guidance system works

**Goal:** steer a frozen ProGen2 model toward stable proteins by training an additional GFlowNet guidance head.

### 1. ProGen2

Provides sequence probabilities and prefix representations. Its weights stay frozen.

### 2. Stability predictor

Scores the whole protein. Higher predicted stability increases its target sampling weight.

### 3. Guidance head

Learns token-level corrections using trajectory balance, to sample the target distribution.

- ProGen2 and a standalone GFlowNet were working independently.
- Integration passed a 256-sequence synthetic test: distribution distance about 0.04–0.05 across three seeds (zero means a perfect match).

# The first stability result: accurate sampling, no gain

**Task:** Mega-scale stability data for one 72-residue protein, 1O6X. 950 training sequences, 204 validation sequences, 204 evaluation variants. Evaluation variants are single amino-acid substitutions.

| | Expected stability (kcal/mol) | |
|---|---|---|
| Distribution | Predicted | Measured |
| Constrained ProGen | 3.894 | 4.280 |
| Exact predicted-reward target | 3.973 | 4.270 |
| Trained GFlowNet (three-seed mean) | 3.973 | 4.270 |

Probability-weighted averages over the same 204 variants; higher is better. "Exact" means probabilities are computed directly, without sampling.

The GFlowNet learned its target accurately. The target itself did not improve measured stability, so we investigated the predictor and reward.

# This week: test the predictor and inspect the reward

## A fair predictor comparison

- Constant training-mean control.
- Ridge: a regularized linear model.
- MLP: a small neural network.

Same frozen ProGen features and partitions. Train on **40, 190 or 950** examples; three seeds with matched subsets. Each subset includes the reference. Choose predictor settings using validation error only.

## A focused reward audit

- Reconstruct the original score and check normalization.
- Test guidance strength $\beta = 0, 0.5, 1, 2$.
- Compare the original reward with a reward capped at predicted reference stability.

$\beta$: guidance strength. Zero recovers ProGen; one is the original setting.

Evaluate every target directly before another GFlowNet fit. The original reward arithmetic checked out.

# The linear predictor gives a better target on this split

**More data improves prediction**

| Train size | Ridge | MLP |
|---|---|---|
| 40 | 1.001 | 0.982 |
| 190 | 0.738 | 0.712 |
| 950 | 0.524 | 0.529 |

Evaluation root mean squared error, in kcal/mol; lower is better. Three-seed means. Constant control: about 1.16.

**Full-data targets, original strength**

| | Measured | Change |
|---|---|---|
| ProGen | 4.280 | — |
| Ridge target | **4.320** | **+0.040** |
| MLP target | 4.261 | −0.019 |

Expected stability in kcal/mol, with $\beta = 1$ and 950 training examples. The MLP loses measured utility in all three seeds.

Similar overall prediction errors can produce different guidance outcomes. These are exact target results; no new GFlowNet was trained this week.

# Why: errors on likely variants matter most

ProGen puts **91.4% of its probability on five variants**. Errors in this small group can dominate the effect of guidance.

| Variant | ProGen probability | Measured | Ridge prediction | MLP prediction |
|---|---|---|---|---|
| V1R | 33.9% | 4.438 | 4.684 | 4.183 |
| G2I | 10.4% | 3.830 | 3.904 | 4.816 |
| D3G | 4.4% | 4.419 | 4.542 | 3.157 |

Full-data seed-0 models; kcal/mol. G2I: glycine at position 2 replaced by isoleucine (same naming convention for other variants).

- The neural model ranks G2I above both more-stable alternatives.
- All three neural models beat ridge on validation error, but produce worse targets.

Evaluate prediction errors where ProGen puts its probability.

# What we learned, and the next focused step

- **The linear baseline is competitive.** At full data, its target gains about 0.040 kcal/mol over ProGen on this split.
- **Reward capping is not a complete fix.** For the neural predictor, it slightly raises mean stability, but lowers the chance of reaching reference-level stability.
- **The limitation is still local evidence.** One protein and an already inspected evaluation set; repeated full-data ridge fits are not independent replications.

## Proposed next step

Fix ridge and the original guidance strength; check additional predefined splits of this protein's data against the neural comparator. Measure overall and ProGen-weighted prediction errors, plus exact target quality. If the gain persists, test whether GFlowNet guidance realizes it. Keep trajectory balance for now: target matching has already worked.